Caleb Lawson
ElectroMechanical Engineer | Robotics | Embedded Systems .
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PROFESSIONAL SUMMARY

Electromechanical engineer focused on autonomous systems, embedded electronics, and unmanned aerial systems. Experience integrating hardware, software, and sensors to develop autonomous and electromechanical systems from concept through testing.

EDUCATION

Wentworth Institute of Technology, Boston, MA
- Master of Science in Electrical & Computer Engineering (ABET Accredited), 08/2026
- Bachelor of Science in Electromechanical Engineering (ABET Accredited), 08/2025

SKILLS

Systems Engineering: Control Systems, Feedback Systems, Electromechanical Design, Sensors & Instrumentation.
Software: SolidWorks (Certified Associate), Microsoft Office (Word, Excel, PowerPoint), MATLAB, Simulink, AutoCAD, Revit.
Photogrammetry Tools; Programming Languages: General Programming, Arduino, Python, C.
Devices: Digital Multimeter, Oscilloscope, Arduino, Power Supply, Drones (Licensed); Technical: 3D printing, soldering.
Professional: Project Management, Communication, Public Speaking, Adaptability, Organization, Conflict Management.

PROJECTS

Autonomous Wildfire Detection Drone System **|** Senior Capstone 01/2025-08/2025
- Designed an embedded drone system integrating onboard sensors, computer vision, and microcontroller logic for real time wildfire detection and geospatial boundary mapping.
- Developed predictive fire spread modeling using wind, humidity, topography, and historical burn data.
- Performed hardware/software integration, signal acquisition, and testing to ensure stable and reliable embedded operation.

Linear Actuating Pillow **|** Intro to Engineering Design 01/2022 - 03/2022
- Collaborated with team members to design a therapeutic back pillow for handicapped persons.
- Designed wiring diagram and embedded software for linear actuators using Arduino.
- Manufactured 3D objects and soldered components used for assembling the adjustable back pads.
- Debugged and troubleshot the hardware and software integration issues.

PUBLICATION

. Kozak, M. Paul, C. Lawson and J. McCusker, "The Design and Development of a Machine Learning Wildfire UAV Swarm Algorithm: IPCA," *2026 IEEE/SICE International Symposium on System Integration (SII)*, Cancun, Mexico, 2026

RELEVANT EXPERIENCE

Electrical Engineering Graduate – Designer 1 | IMEG, Boston, MA 08/2025-Present
- Performed system level electrical analysis, redundancy planning, and safety critical design using NEC/NFPA standards.
- Produced detailed wiring diagrams, load paths, and electrical architecture documentation similar to capture workflows.
- Collaborated with multidisciplinary engineering teams and contractors to deliver efficient, reliable, and cost-effective building system solutions, developing strong systems and integration experience.

Project Controls Intern | Walt Disney Imagineering, Orlando, FL 08/2024-01/2025
- Evaluated design documentation and provided trend analyses to compare budget estimates against current project costs.
- Led meetings to identify risks to project timeline and costs, handling all risks under $50k.
- Collaborated with cross-functional teams, including project managers and contractors, to achieve project objectives.
- Applied Excel skills to efficiently learn and utilize new software, to create interactive dashboards for improved oversight.

Engineering Intern | Vanderweil Engineering, Boston, MA 01/2024-04/2024
- Applied skills in 3D modeling using SolidWorks to create detailed representations of electrical components.
- Performed Value Engineering to showcase sustainability in new design, making the extra cost worth it.
- Conducted analysis of power systems, using software such as Pipe Flo, AutoCAD, and Revit.
- Generated detailed drawings and engineering reports, documenting power system designs.

Assistant Project Manager | JM Electrical, Lynnfield, MA 05/2023 - 08/2023
- Obtained drone license, enabling cost-effective and expedited site surveys using drones.
- Utilized imagery program for photogrammetry for the creation of 3D site models.
- Utilized advanced project management software to accurately estimate manpower requirements.
- Consistently provided swift and high-quality support to upper management.

MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE), Wentworth Alumni Association